NLP.

Experiment - 01.

- **Aim:** Perform Tokenization (Whitespace, punctuation based, Treebank, Tweet, MWE) using NLTK library. Use porter stemmer and snowball stemmer for stemming. Use any technique for lemmatization.
  Input Dataset - use any sample sentence.

- THEORY:

(1) Introduction -

- Text preprocessing is a fundamental step in Natural Language processing (NLP). Raw Textual data contains noise such as punctuation, casing differences, inflections and stopwords.
- Before applying machine Learning or deep learning algorithms, the text must be normalised.
- Major Techniques include -
  - Tokenization.
  - Stemming.
  - Lemmatization.

(2). Tokenization -

2.1. Tokenization is a process of breaking a sequence of text into smaller units called tokens.
Mathematically, let a document be represented as -

$$D = \{ w_1, w_2, w_3, \ldots w_n \}$$

Tokenization is a transformation function:

$$T(D) = \{ t_1, t_2, t_3 \ldots, t_k \}$$

Where,
- D = original document.
- T = tokenization function.
- t_i = tokens.

2.2. Types of Tokenization in NLTK.

(A). Whitespace Tokenization.

Splits text based on spaces.

$$T_{ws}(D) = D.\text{split}(" \quad ")$$

Example : " Natural Language processing is powerful "
Output: ["Natural", "Language", "Processing", "is", "powerful"].
Limitation : Does not remove punctuation.

(B). Punctuation - based Tokenization.

Uses punctuations as delimiters :

$$T_{punc}(D) = \text{Split}(D, \text{punctuations})$$

Example - "Hello, world!"
Output : ["Hello", "world"]

(C) Treebank Tokenization.

Implements Penn Treebank rules :
- Splits contractions.
- Separation punctuation properly.

Example: "Don't worry."
Output: ["Do", "n't", "worry", "."]
It follows linguistic rules for better syntactic representation.

(D). Tweet Tokenization -
Designed for social media text:
- Handles Hashtags.
- Mentions @user.
- Emojis.
- Repeated characters.

(E). Multiword Expression (MWE) Tokenization -
Combines specific word sequences into a single token.
If:

$$M = \{ (w_i , w_{i+1}) \}$$

Then:

$$T_{mwe} (D) = Merge (w_i , w_{i+1})$$

Example:
Machine Learning → Machine-Learning.

(3). Stemming.

3.1. Definition.
Stemming reduces words to their root form by removing suffixes.

Mathematically, $S(w) = r$.

where,
- w = orignial Word.
- r = root word.

Example -
playing → play.
connected → connect.

3.2. **Porter stemmer.**
Developed by Martin Porter (1980): It applies rule based suffix stripping:
Example rule:
If word ends with "ing" ⇒ remove "ing"
playing → play
studies → studi.
It is aggressive and may produce non-dictionary roots.

3.3. **Snowball Stemmer.**

Improved version of Porter Stemmer
$S(w) = r$

Features -
- Faster.
- Supports multiple languages.
- More consistent results.

Example - better → better.
running → run.

(4). **Lemmatization**

4.1. **Definition.** - Lemmatization reduces a word to its dictionary base form (lemma) using vocabulary and morphological analysis.

Mathematically, $L(w, pos) = lemma.$

where,
- w = word.
- pos = part of speech.
- lemma = base form.

Example - am → be.
running → run
better (adj) → good.

Unlike stemming, lemmatization returns meaningful dictionary words.

(5) Sample sentence used.

"Machine learning models are running efficiently and researcher are analyzing tweets about artificial intelligence".

(6). Comparative Analysis -

	Technique.	Output Quality	Dictionary Word?	Speed.	Use Case.
1.	Whitespace tokenizer.	Basic.	Yes.	Fast.	Simple tasks.
2.	Punctuation Tokenizer.	Clean words	Yes.	Fast	Basic NLP.
3.	Treebank Tokenizer.	linguistically accurate.	Yes.	Moderate	NLP pipelines.
4.	Tweet Tokenizer.	Social media ready.	Yes.	Moderate	Twitter / Instagram data.

	Technique.	Output Quality	Dictionary word.	Speed.	Use Case.
5.	MWE Tokenizer	Combines phrases	Yes.	Moderate.	Domain specific phrases.
6.	Porter Stemmer	Rough root.	No.	Fast	IR Systems.
7.	Snowball Stemmer.	Better root	Sometimes	Fast	General NLP.
8.	Lemmatization	Accurate Base Form.	Yes.	Slower	ML/DL tasks.

* CONCLUSION: Text preprocessing is a critical step in NLP. Tokenization transforms text into manageable units, stemming reduces inflected words to root forms using heuristic rules and lemmatization converts words to meaningful base forms using morphological analysis.

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